1.6 Graphs of Other Trigonometric Functions

Name _____ Period ____ Date _____

Pre-Calculus 1.6 · Student sheet

I can…

- find the period and asymptotes of a tangent or cotangent function.
- sketch tangent and cotangent graphs.
- sketch secant and cosecant graphs using their reciprocal functions.

Vocabulary · fill in the missing word

Vertical asymptote — a vertical line x = a that a graph _____ but never reaches

Reciprocal function — csc x = _____ = 1/cos x

Part 1 · Where is tangent undefined?

tan x = sin x / cos x

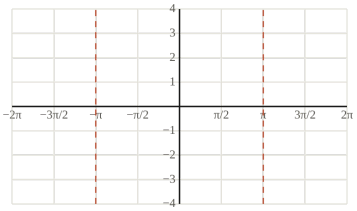
x	0	π/4	π/2	3π/4	π
sin x					
cos x					
tan x					

Part 2 · Notes and examples (fill in each note, then work the example)

Tangent: y = a tan bx has period _____; asymptotes where bx = ±π/2.

Example 1

Sketch $y = \tan(\frac{x}{2})$. Find the period and two consecutive asymptotes.



Cotangent: y = a cot bx has period π/b; asymptotes where bx = 0, _____.

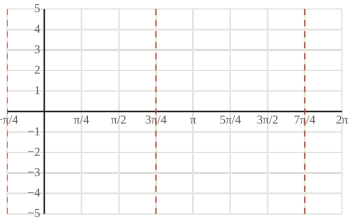
Example 2

Find the period and asymptotes of $y = 2\cot(\frac{x}{3})$ on one cycle.

Cosecant and secant: asymptotes where the sine (or cosine) is _____; the graphs touch at its maxima and minima.

Example 3

Sketch $y = 2\csc(x + \frac{\pi}{4})$ using $y = 2\sin(x + \frac{\pi}{4})$. Where are the asymptotes?



Example 4

Find the period and the asymptotes on $[0, 2\pi]$ of $y = \sec 2x$.

Part 3 · Partner practice

Partner 1

Find the period and two consecutive asymptotes of $y = \tan 2x$.

Partner 2

Find the period of $y = 3 \csc 4x$.